StockLens - Scan your spending, See your missed investing
Design Proposal

App Concept

App Name: StockLens

Overview: StockLens is a financial literacy and investing-awareness app that transforms everyday spending into tangible investment lessons. By scanning shopping receipts with their phone's camera, users instantly see how their purchases could have grown if invested (in the past and future) in high-performing stocks, turning abstract financial concepts and ideas into personal, visual, and emotionally engaging insights.

Rather than theoretical lessons or static calculators, StockLens offers a personal, visual view of the opportunity cost that comes with spending and not investing. For example: "That £40 takeaway could have grown to £250 over the next five years, if invested today" or "That £40 takeaway if invested 10 years ago would now be worth £450." Historical performance data, future projections, and gamified insights make financial education interactive, relatable, and emotionally engaging.

Primary Objectives:

1. **Behavioural Change:** Shift users from passive spending to conscious investing by making opportunity costs tangible and emotionally resonant.
2. **Financial Literacy:** Educate users on compound growth, stock market potential, and investment fundamentals through real-world examples from their own spending.
3. **Action Conversion:** Motivate users to open investment accounts and begin building wealth, bridging the gap between awareness and action.

The Problem: Traditional finance apps focus on budgeting (tracking what you've spent) or require tedious manual data entry. They fail to connect everyday purchases with investment opportunities, leaving users unmotivated and disengaged. Most young adults know they "should" invest but can't visualise the real impact of redirecting small amounts of money every so often.

The Solution: StockLens uses OCR technology to scan receipts instantly, converting purchase amounts into "missed investment opportunities." Users see concrete possible historical returns: "that £40 takeaway could have been worth £250 in 5 years if invested in Apple" and theoretical projections "your £80 trainers could grow to £200 over 10 years."

Key Innovation: Unlike any budgeting or investment app, StockLens combines three elements: (1) receipt scanning for real-world data capture, (2) historical stock performance analysis showing actual missed opportunities, and (3) forward-looking projections that visualise future wealth potential. The app builds a personalised "wealth timeline" dashboard and summary/analytics page, gamifying financial awareness and creating emotional investment in behaviour change.

Example Scenario: Amir, a 22-year-old student, scans his £80 trainer receipt. StockLens reveals that amount invested in NVIDIA five years ago would now be £480. Shocked by the tangible loss, he begins scanning all purchases. After two months of tracking £600 in "missed opportunities," Amir opens a Trading 212 account and makes his first investment, awareness becomes action.

Target Users

Primary Users: Financially illiterate/curious young adults (ages 18-28)
StockLens targets digitally-native young adults in the UK who are earning or managing their own money for the first time but lack the confidence, knowledge, or motivation to start investing.

This segment represents approximately 6.8 million people aged 18-28 in the UK, with an estimated 3.2 million actively seeking financial education tools.

Primary users include university students, recent graduates, and early-career professionals who are comfortable with technology but intimidated by traditional finance platforms.

Demographics:

- Age: 18-28 years old
- Location: UK-based
- Occupation: Students, part-time workers, entry-level professionals (retail, hospitality, tech, healthcare, etc)
- Income: £0-£40,000 annually; limited disposable income but consistent small purchases (avg. £150-400/month on non-essentials)
- Tech Profile: iPhone and Android users; heavy mobile app usage

Psychographics:

- **Attitudes**: Financially curious but overwhelmed; aware they "should" invest but paralysed by complexity and fear of "doing it wrong."
- **Values**: Prioritise instant gratification over delayed rewards but aspire to financial security; value visual, relatable learning over textbook theory or financial jargon.
- **Lifestyle**: Mobile-first, social media-driven (TikTok, Instagram, YouTube); influenced by finance creators and peer behaviour; impulse spenders who experience post-purchase guilt
- **Current Apps Used**: Monzo, Revolut, Starling Bank for spending tracking; rarely engage with investment platforms (Trading 212, InvestEngine) despite awareness.

Pain Points:

- Feel they've "missed the boat" on major stock gains (Tesla, Apple, NVIDIA); demotivated by past inaction
- Frustrated that existing budgeting apps (Emma) only show restrictions and "what you've lost," not growth opportunities
- Intimidated by investing jargon, market volatility, and fear of making wrong choices
- Cannot visualize long-term impact of small, daily spending decisions
- Lack structured financial education; rely on fragmented social media content

Needs:

- **Frictionless data capture**: No manual entry required, use camera/OCR to scan receipts instantly, manual entry only required if OCR fails
- **Personal, relatable examples**: Must show their actual spending converted to investment opportunities, not generic scenarios
- **Visual, emotional impact**: Need "shock value" calculations that create immediate emotional response (e.g., "Your £40 takeaway = £250 in 10 years")
- **Jargon-free education**: Plain language explanations of compound growth, opportunity cost, and investing basics

- **Instant gratification**: Immediate insights and feedback; cannot wait for weekly/monthly reports
- **Low commitment**: Must work without requiring investment account setup initially; education before action
- **Trust and transparency**: Clear data sourcing (which stocks, why these calculations, historical accuracy)
- **Gamification**: Progress tracking, milestones, and visual growth charts to maintain engagement
- **Actionable next steps**: Clear pathway from "awareness" to "opening first investment account"

Goals:

- Understand how small, regular investments compound over time
- Build the confidence to open first investment account (Trading 212, InvestEngine, etc.)
- Reduce unnecessary spending by visualising its opportunity cost
- Learn investing fundamentals through real-world examples, not theory

Secondary Users: Parents and Educators (ages 35-50)
Secondary users include parents, educators, and mentors who want to introduce financial literacy concepts in a practical way.

Teachers can use StockLens in classrooms to demonstrate the long-term impact of spending versus investing.

Parents can use it at home with children to spark conversations about money management.

This segment includes approximately 500,000 secondary school teachers in the UK and 8 million parents with teenage children (ages 13-17)

Demographics:

- **Age**: 35-50 years old
- **Location:** UK-based
- **Occupation:** Secondary school teachers (business studies, economics, maths) and parents with teenage children
- **Income:** £30,000-£55,000 annually (teachers); £30,000-£70,000 (parents in professional roles)
- **Tech Profile:** Moderate app users; prefer simple, intuitive interfaces; IOS or Android users, limited engagement with complex fintech tools

Pain Points:

- Struggle to engage teenagers and young adults in financial literacy conversations, abstract concepts don't resonate.
- Traditional finance education materials (textbooks, videos) feel disconnected from students' real lives.
- Teenagers dismiss parental/teacher advice about money as "boring" or "not relevant to them."
- Lack practical, age-appropriate tools that bridge theory and real-world application.
- Difficulty demonstrating long-term consequences of financial decisions in ways young people can visualise.
- Concerned that students/children are entering adulthood without basic investing knowledge.

Needs:

- Practical teaching tool that transforms abstract concepts (opportunity cost, compound growth) into tangible, visual lessons
- Age-appropriate interface suitable for 13–17-year-olds without requiring personal investment accounts
- Ability to use real family/classroom spending data as discussion starting points
- Simple demonstration mode that doesn't require extensive app training or financial expertise
- Curriculum-aligned content for UK secondary education (Key Stage 4-5)

Goals:

1. Use StockLens in classroom settings to demonstrate real-world financial decision-making during business studies or maths lessons
2. Scan grocery receipts, family purchases, or hypothetical spending with children at home to spark money conversations
3. Help teenagers understand that "small amounts matter" before they develop poor spending habits
4. Prepare 16–17-year-olds for financial independence and investing before they turn 18
5. Create "lightbulb moments" where young people emotionally connect their spending choices to future wealth

Use Cases:

- **Classroom Integration:** Teachers project app during lessons, scanning example receipts to show class how spending translates to missed investment opportunities.
- **Homework Activities:** Students scan family receipts and calculate opportunity costs as part of business studies coursework.
- **Family Financial Education:** Parents scan weekly grocery shop with teenage children, discussing "what if we'd invested this £20 instead?"
- **Pre-Investment Education:** Introduce investing concepts to 16-17-year-olds before they can legally open accounts at 18.

Note: Secondary users are educational adopters, not primary revenue drivers. App design prioritizes primary user experience (18-28-year-olds); secondary features (e.g., "teacher mode," family accounts, or curriculum guides) would be considered for future iterations based on primary user traction.

Non-Target Users:
StockLens intentionally excludes the following segments to maintain focus on its core value proposition:

- **Experienced Investors**: Users actively trading stocks, managing portfolios (£10,000+ invested), or seeking advanced analytics, better served by Trading View, Bloomberg, or brokerage platforms
- **Debt Management Focus**: Individuals in significant debt (credit cards, loans) requiring debt repayment strategies rather than investment education, better served by StepChange, Citizens Advice

- **Pre-Retirement Planning**: Users 50+ seeking pension optimization or retirement income strategies, better served by Nutmeg, PensionBee, or financial advisors
- **Professional Financial Advice Seekers**: Users expecting regulated investment recommendations or portfolio management, StockLens is educational only, not FCA-regulated financial advice

User Personas

The following personas represent the primary and secondary target audiences for StockLens. Each persona includes detailed attributes that inform design decisions, feature prioritisation, and user experience considerations throughout the app development process.

Persona 1: Liam Thompson (Primary User - Early Awareness Stage)

Attribute	**Details**
Name	Liam Thompson
Age	19
Occupation	First-year university student (Business & Economics major)
Location	Manchester, UK
Income	Part-time retail worker, £800/month (~£9,600 annually)
Device	iPhone 14, uses mobile banking (Monzo), PayPal, and basic budgeting apps
Financial Background	Works part-time in retail; spends heavily on food deliveries (£150/month), clothes (£80/month), and gaming subscriptions (£30/month). Aware of investing concepts from university lectures but hasn't started due to lack of practical knowledge and fear of "doing it wrong."
Current Finance Habits	Tracks spending loosely via Monzo banking app, checks balance daily. Understands "opportunity cost" academically but struggles to connect theory to his day-to-day £8 Uber Eats orders and £40 online shopping purchases. Saves sporadically (£50-100/month) without clear investment strategy.
Technology Profile	Very comfortable with scanning QR codes and receipts (uses them for store loyalty programs). Active on TikTok and Instagram where finance influencers appear in his feed. Expects instant, frictionless app experiences; abandons apps requiring manual data entry.
Pain Points	• Doesn't see the long-term impact of his spending habits, can't visualise how his £150/month on Deliveroo could compound if invested. • Frustrated that budgeting apps like Emma only show "what he's lost," not what he could have gained. • Feels overwhelmed by technical investing language on platforms like Trading 212 and InvestEngine • Believes he's "missed the boat" on big tech stocks (NVIDIA, Apple, Tesla) and it's "too late" to start.

	• Experiences post-purchase guilt after impulse spending but no tools to quantify the real cost.
Goals	• Understand the impact of small, regular investments using his actual spending data • Build confidence to open his first investment account within 3-6 months • Reduce food delivery spending by 20% by visualizing opportunity costs • Learn investing fundamentals through relatable examples, not textbook theory
Needs	• **Instant visualization**: Scan receipt → see missed opportunity calculation immediately. • **Relatable benchmarks**: Show growth using popular stocks he's heard of (Apple, NVIDIA, Amazon, etc). • **Jargon-free education**: Plain language explanations of compound growth and opportunity cost.

Persona 2: Maya Patel (Primary User - Active Learning Stage)

Attribute	**Details**
Name	Maya Patel
Age	22
Occupation	Final-year university student (Psychology major), part-time barista
Location	Edinburgh, UK
Income	£1,100/month from part-time work (~£13,200 annually)
Device	iPhone 15 Pro, uses Monzo for expense categorisation, experimenting with Revolut and Chip
Financial Background	Budget-conscious; already saves £100 monthly into a savings account but struggles with discipline when social events arise (nights out, concerts, dining). Has basic awareness of stock markets from TikTok finance creators and YouTube videos but still hasn't acted.
Current Finance Habits	Uses Monzo to categorise expenses meticulously (groceries, entertainment, transport). Tries to save £100 monthly but often spends it on social activities. Checks her savings balance weekly. Wants to "do more" with her money but finds traditional investing platforms intimidating.
Technology Profile	Loves experimenting with new productivity and finance apps (has downloaded 15+ in the past year). Comfortable with OCR scanning apps (uses them for receipts, documents). Active digital wallet user (Apple Pay). Expects seamless UX, abandons apps with clunky interfaces.
Pain Points	• Finds traditional investing platforms (Freetrade, InvestEngine) intimidating and full of jargon • Can't visualize how her spending choices (£50 brunch, £40 concert tickets) affects long-term wealth

	• Feels she's missing opportunities compared to peers who already invest, FOMO drives anxiety • Budgeting apps show her spending categories but don't connect them to future wealth potential • Wants accountability without judgment, dislikes apps that feel "preachy" about spending
Goals	• Reduce unnecessary spending by 10-15% and redirect savings into investments • Learn through relatable, real-world examples rather than abstract theory • Build a diversified investment portfolio starting with £50-100/month • Understand stock market basics (compound growth, diversification) through her own data
Needs	• Personal accountability: Dashboard showing cumulative missed opportunities to create urgency • Forward-looking projections: Show how redirecting £50/week could grow to £5,000 in 5 years • Low-pressure education: Learn at her own pace without feeling overwhelmed • Progress tracking: Visual milestones celebrating financial behaviour change

Persona 3: Jordan Kim (Primary User - Hesitant Investor)

Attribute	**Details**
Name	Jordan Kim
Age	25
Occupation	Entry-level software developer
Location	London, UK
Income	£32,000 annually (£2,200/month after tax)
Device	Android (Samsung Galaxy S23), power user of fintech apps (Revolut, Binance, Robinhood)
Financial Background	Recently started full-time job after graduation. Still adjusting from student life; spends heavily on gadgets (£200-400/month), eating out (£300/month), and tech subscriptions (£50/month). Has dabbled in cryptocurrency via Binance but lost £200, which reinforced fear of investing. No traditional stock investments yet.
Current Finance Habits	Uses Revolut for budgeting and tracks crypto holdings daily (though inactive). Reads about stock markets on Reddit (r/UKPersonalFinance) but never commits due to "fear of losing money." Considers himself financially savvy but paralyzed by analysis paralysis.
Technology Profile	Android power user, comfortable with complex apps and APIs. Uses automation tools (IFTTT, Tasker). Expects high-quality UX design, judges apps harshly. Scans receipts occasionally for expense tracking but doesn't use OCR regularly.

Pain Points	• Overwhelmed by market fluctuations and financial news, feels he needs "perfect timing" to invest • Thinks budgeting apps are too focused on restrictions ("stop spending") rather than opportunities ("start building wealth")Doesn't trust himself to pick the right stocks, fears making wrong choices • Crypto losses created anxiety around all investing; needs reassurance that small, diversified investments are safer • Knows he's overspending on non-essentials but can't emotionally connect spending to long-term consequences
Goals	• Understand long-term benefits of investing small amounts instead of overspending on short-term luxuries • Build confidence in stock market investing through simulations and projections • Redirect £300/month from gadgets/takeaway into diversified investments • Learn risk management and diversification strategies through practical examples
Needs	• Reassurance through data: Show that consistent small investments outperform "perfect timing" • Risk education: Explain volatility vs long-term growth using historical data • Simulations: "What if I'd invested £X every month for 5-10+ years?" projections

Persona 4: Aisha Okonkwo (Primary User - Financial Stability Seeker)

Attribute	**Details**
Name	Aisha Okonkwo
Age	28
Occupation	Nurse, working shifts (NHS)
Location	Birmingham, UK
Income	£28,000 annually (~£1,900/month after tax)
Device	iPhone 13, active on WhatsApp and Instagram, less active on fintech apps
Financial Background	First in family to work in healthcare. Has steady income but limited financial literacy, never taught about investing growing up. Family relies on her for financial support (sends £150/month to parents). Pays bills on time but little left for saving after expenses and family support.
Current Finance Habits	Uses cash frequently and keeps paper receipts for budgeting (old-school habit from parents). Doesn't use budgeting apps, finds them overwhelming. Checks bank balance weekly. Wants to save but feels guilty spending on non-essentials when family needs support.

Technology Profile	Moderate app user, prefers simple, intuitive interfaces. Not familiar with fintech apps beyond mobile banking (Barclays app). Comfortable scanning receipts for loyalty programs (Tesco Clubcard). Uses WhatsApp and Instagram daily but not TikTok.
Pain Points	• Investing feels inaccessible and intimidating, believes it's "for rich people" or requires large sums • Feels guilty when money is spent on non-essentials instead of saving or supporting family • Doesn't relate to complex graphs, charts, or trading apps, needs visual simplicity • Lacks financial role models or mentors to guide investing decisions • Worried about making mistakes with limited disposable income
Goals	• Build long-term savings by learning how small investments grow over time • Provide financial stability for younger siblings through smart money management • Understand investing basics in clear, non-technical language
Needs	• Simple visual evidence: "Small changes now = big differences later" messaging • Low barrier to entry: Show that £20/month is valuable, not dismissing small amounts • Trust-building: Transparent calculations and data sources to overcome scepticism

Persona 5: David Miller (Secondary User - Educator/Parent)

Attribute	**Details**
Name	David Miller
Age	42
Occupation	Secondary school teacher (Maths & Business Studies)
Location	Glasgow, UK
Income	£38,000 annually
Device	Google Pixel 9, moderate app user, prefers simple interfaces
Financial Background	Steady middle-income, maintains savings account (£8,000) but doesn't actively invest beyond workplace pension. Wants to pass financial literacy to his two teenage kids (ages 15 and 17). Comfortable with personal finance but lacks tools to engage young people effectively.
Current Finance Habits	Uses Excel spreadsheets for budgeting (old-school approach). Keeps paper receipts and reviews monthly spending manually. Not an active user of investing apps, views them as "too complicated" for teaching purposes. Saves £300/month into ISA.

Technology Profile	Moderate with apps, prefers simple, intuitive interfaces. Uses Google Classroom for teaching. Comfortable with basic tech but not a power user. Sceptical of overly complex apps.
Pain Points	• Notices his students (and own kids) lack financial awareness, they don't understand long-term consequences of spending • Feels most finance apps are "too grown up" for teens or require investment accounts they can't open yet • Finds it hard to engage kids in long-term thinking about money, abstract lessons don't resonate • Struggles to connect classroom theory (compound interest formulas) to real-life examples students care about • Wants tools that make financial literacy "cool" and relevant to teenagers' lives
Goals	• Use StockLens as a teaching tool in Year 10/11 Business Studies classes to demonstrate opportunity cost and compound growth • Encourage his own children (ages 15, 17) to start thinking about investing before they turn 18 • Create "lightbulb moments" where students emotionally connect spending to future wealth • Integrate practical financial literacy into curriculum-aligned lessons
Needs	• Classroom demonstration mode: Project app on whiteboard, scan example receipts, show class calculations in real-time • Age-appropriate content: Suitable for 14-17-year-olds without requiring personal accounts

Persona Summary & Design Implications

Design Consideration	Personas Addressed	Implementation in StockLens
Frictionless OCR scanning	All primary users	Camera-based receipt scanning with instant results, no manual entry
Jargon-free language	Liam, Aisha, David	Plain language throughout; tooltips explain financial terms simply
Visual "shock value"	All primary users	Large, bold numbers showing missed opportunities (e.g., "£40 → £250 in 5 years")
Gamification & progress	Liam, Maya, Jordan	Dashboard with milestones, wealth timeline, and streak tracking
Low barrier to entry	Aisha, Liam, Maya	Emphasize value of small amounts (£20-50/month)
Peak usage optimization	All primary users	Quick-scan functionality for evening/commute/lunch usage patterns
Data analytics	Maya, Jordan	Detailed spending patterns, number of scans, total missed opportunity using all scans, etc.

Technical depth	Jordan	Optional advanced view showing calculations and data sources
Platform compatibility	All users	iOS and Android support (via React Native)
Historical + future projections	All primary users	Toggle between "what you've missed" and "what you could gain"
Trust & transparency	Aisha, Jordan, David	Clear data sourcing and calculation explanations on every screen
Instant feedback	All primary users	Results in <2 seconds; real-time dashboard updates

Minimum Viable Product (MVP) Features

Essential Features (MVP)

The following features represent the core functionality required for StockLens to deliver its primary value proposition: transforming everyday spending into tangible investment lessons. These features directly address the critical needs identified across all user personas and align with the user flow diagram and app map structure.

Feature	Description	Target User Need Addressed	Technical Implementation
Sign Up & Login	User account creation with email/password; returning user login with credentials.	Secure data storage; personalized experience (all users)	Firebase Authentication; email validation; password encryption.
Face ID/Touch ID Authentication	Biometric authentication option for quick, secure login on supported devices.	Frictionless login: enhanced security (Maya, Jordan prefer convenience)	iOS: Local Authentication framework; Android: BiometricPrompt API.
Dashboard - Populated State	Displays list of scanned receipts with date, amount, and potential investment value; summary statistics at top.	Visualises spending patterns; tracks cumulative missed opportunities (all primary users)	local database query; calculated summary metrics
Camera Screen	Native camera interface optimized for receipt capture; guidelines overlay to assist framing.	Quick, intuitive receipt capture (all users expect frictionless experience)	React Native Camera library; camera permissions handling

OCR Processing	Extracts receipt total, merchant name, and date from photographed receipt using optical character recognition.	Eliminates manual entry; instant data extraction (critical for Liam, Maya who abandon manual-entry apps)	OCR Space API; text recognition algorithm; currency/number parsing
Manual Entry (Fallback)	Modal input form for manual amount entry when OCR fails or user prefers manual input.	Ensures 100% usability even with poor-quality receipts (Aisha uses cash receipts, thermal paper)	Modal form with numeric keyboard; currency formatting; validation
Confirmation Screen	Review screen showing extracted amount with "Is This Correct?" prompt and edit option before projection.	Builds trust; prevents errors; allows correction (Jordan needs reassurance about accuracy)	Display parsed data; edit button triggers manual entry modal; confirm/rescan actions
Investment Projection Calculator	Core algorithm converting receipt amount into historical growth (1-20 years ago) and future projections (1-20 years) using select stocks.	Demonstrates missed opportunities; teaches compound growth (addresses all personas' primary pain point)	Financial data API (Alpha Vantage); compound interest formula; historical price data; projection algorithm
Results Screen	Displays "shock value" visualisation: original amount vs. historical value vs. future projections with large, bold numbers and simple growth chart.	Creates emotional impact; makes opportunity cost tangible (Liam's "lightbulb moment," Aisha's visual learning)	Chart library (react-native-chart-kit); bold typography; color-coded gains; stock logo/name display
Save Receipt	Saves scanned receipt with projections to user's personal history stored locally on device.	Builds historical record; enables progress tracking (Maya wants data-driven accountability)	Local database (SQLite) no cloud database.
Receipt Details View	Detailed view of individual saved receipt showing full projection data, merchant, date, and delete option.	Allows review of past scans; provides context for spending patterns (all users)	Navigation to detail screen; formatted data display; delete trigger

Delete Receipt Confirmation	Modal confirmation preventing accidental deletion: "Are you sure? This cannot be undone."	Prevents data loss; builds user trust (all users)	Modal dialog with Yes/No actions; delete from local database on confirm
Summary/Analytics Screen	Displays aggregate statistics: total spending scanned, total missed investment value, top spending categories (if categorisation implemented), trends.	Provides overview; motivates behaviour change (Maya, Jordan want analytical insights)	Data aggregation queries; summary calculations; basic chart visualizations
Settings Screen	Hub for account management, preferences, app information, and data controls.	Provides user control; transparency (all users need account management)	Settings menu with navigation to sub-sections; profile display; app version info
Log Out Confirmation	Modal confirmation before logging out: "Log Out?" with Yes/No options.	Prevents accidental logout; maintains session (all users)	Clear auth token on confirm; return to login screen; confirm/cancel actions
Clear Data Warning	Modal warning before clearing all saved receipts: "This will delete all your data and account, Cannot be undone."	Prevents accidental data loss; gives users control to reset (all users)	Delete all local database records on confirm; reset to empty state; Yes/No actions
Error Handling	User-friendly error messages for common failures: OCR failure, network errors, invalid inputs.	Maintains usability during failures; guides users to solutions (Aisha needs clear guidance, not technical errors)	Try-catch blocks; error state UI; retry buttons; fallback to manual entry
Offline Functionality	Core scanning and projection features work without internet; sync calculations when online.	Ensures reliability; accommodates poor connectivity (all users expect app to "just work")	Local calculation fallback using cached stock data; queue for API calls when offline

Nice-to-Have Features (Future Versions)

The following features enhance StockLens beyond the MVP, adding depth, engagement, and personalization. These are prioritized based on user value, technical complexity, and development timeline constraints.

Feature	Description	Persona Benefits	Technical Requirements	Priority Level	Rationale
Receipt Image Storage	Store actual receipt photos alongside data for reference/proof.	All users: verification; tax purposes (future B2B pivot); builds trust	Image storage (local or cloud); compression; retrieval system; storage management	High	Nice to have but increases storage requirements; not critical for core value prop; privacy concerns
Social Sharing	Share receipt projections as formatted graphics to Instagram Stories, TikTok, or via link. Privacy controls hide personal data.	Liam and Maya: validation and peer influence; drives organic growth via social proof	Image generation library (react-native-view-shot); share API; privacy toggle; branded graphics templates	High	Directly requested in personas; low technical complexity; high viral potential for user acquisition
Greater Stock Comparison	Allow users to view 10+ stocks for comparison in a single view.	All primary users: personalised scenarios; Jordan: analytical depth; diverse investment education	Multiple API calls per scan; UI redesign for multi-stock display; stock selection interface	High	High user value; moderate complexity; enables more personalized learning
Educational Tips & Tooltips	Contextual financial literacy tips appear throughout app (e.g., explaining compound interest, diversification).	Liam, Aisha: jargon-free education; David: teaching aid; builds confidence	Content management system for tips; conditional display logic; progress tracking (tips viewed)	High	Enhances education goal; low complexity; requires content creation

Receipt Auto-Categorisation	AI classifies receipts into categories (groceries, dining, transport, entertainment); shows category-based missed opportunities.	All users: understand which spending categories drive highest opportunity costs; Maya: analytical depth	ML classification model; OCR data pipeline enhancement; category taxonomy; training data	Medium	Adds analytical value; high complexity; not critical for core value prop
Cloud Sync Across Devices	Sync receipt history and projections across user's devices (iPhone + iPad, multiple Android devices).	Maya, Jordan: multi-device users; seamless experience; data backup	Cloud database (Firebase Firestore); authentication sync; conflict resolution	Medium	High user value for multi-device users; moderate complexity; infrastructure costs
Advanced Analytics Dashboard	Deeper insights: spending trends over time, category breakdowns, day-of-week patterns, monthly comparisons.	Maya, Jordan: data-driven personalities; David: classroom analysis tool	Enhanced data queries; advanced charting library; statistical calculations; date range filters	Medium	Appeals to subset of users; moderate complexity; MVP analytics sufficient for most
Gamification & Challenges	Badges for milestones (10/50/100 receipts scanned); streak tracking; optional leaderboards; family/friend challenges.	Liam: competitive motivation; Maya: progress celebration; David: classroom engagement tool	Game logic system; badge asset creation; leaderboard database (if competitive); push notifications	Low	Engagement booster; moderate complexity; risk of feeling gimmicky if poorly executed

APIs

MVP Features - Essential APIs

API	Functionality Enabled	Specific Use Cases	Pricing/Cost
Firebase Authentication	• User account creation • Secure login/logout • Password reset • Session management • Email verification	• Liam creates account with email/password • Maya logs in on new device • Jordan resets forgotten password • Secure credential storage	Free (up to 50K MAU)
OCR Space API	• Text detection from images • Amount recognition	• Camera captures receipt photo → extracts £40 total	Free
Alpha Vantage API	• Historical stock price retrieval • Current stock prices • Daily/monthly time series data • Multiple stock ticker support	• Fetch NVIDIA price from 5 years ago • Calculate £40 → £250 historical projection • Retrieve Apple, S&P 500, Tesla historical data • Support 1Y/3Y/5Y/10Y/20Y timeframes	Free up to 500 requests per day
SQLite	• Local receipt history storage • User preferences persistence • Cached stock price storage • Offline data access • Dashboard metrics calculation	• Store 23 scanned receipts for Maya's dashboard • Save user's dark mode preference • Cache stock prices for offline projections • Calculate "Total Possible Value" metrics	Free (built-in)
React Native Camera	• Camera viewfinder display • Photo capture • Image quality adjustment	• Open camera for receipt scanning • Capture receipt photo with green guide overlay • Optimize image for OCR processing	Free (native)
Local Authentication (Biometric)	• Face ID authentication • Touch ID authentication • Fallback to passcode	• Jordan uses Face ID for quick login • Aisha unlocks app with fingerprint • Secure access without password typing • iOS/Android cross-platform support	Free (native)

Nice-to-Have Features - Additional APIs

API	Functionality Enabled	Specific Use Cases	Pricing/Cost
Firebase Firestore	• Cloud data synchronization • Cross-device receipt access• Real-time updates • Automatic backup • Conflict resolution	• Maya scans on iPhone, views on iPad • Jordan's receipts sync between devices• Automatic cloud backup of scan history • Real-time dashboard updates	Free (50K reads/20K writes daily)
Firebase Storage	• Receipt image storage • Photo backup • Image retrieval • CDN delivery• Secure file access	• Store actual receipt photos for verification • Backup images for tax purposes• Retrieve receipt image in detail view • Future: OCR re-processing capability	5GB free
Native Share API	• Social media sharing • Generate share sheets • Cross-platform sharing • Text/image/URL sharing • App-to-app communication	• Share "£40 → £250" projection to Instagram Stories• Send receipt summary via WhatsApp • Post achievement to TikTok • Copy share link to clipboard	Free (native)
react-native-view-shot	• Screenshot generation • Component-to-image conversion • Branded graphic creation • Custom image rendering • Social media format optimization	• Convert results screen to Instagram-sized image • Generate branded shareable graphic with StockLens logo • Create 9:16 Story format for TikTok • Export projection as PNG for saving	Free (library)
Google Cloud Natural Language	• Text classification • Merchant categorisation • Spending pattern analysis • Entity recognition • Sentiment analysis	• Auto-categorize "Tesco" as Groceries• Classify "Uber Eats" as Dining/Takeaway• Identify "Zara" as Clothing/Retail • Generate category-based analytics	First 5K requests/month free
TensorFlow.js	• On-device ML inference • Custom categorisation models•	• Train custom receipt categorisation model • Process merchant names locally (no cloud)	Free (open-source)

	Privacy-preserving classification • Offline ML capabilities • Transfer learning	• Offline category predictions • Privacy-first alternative to cloud ML	
Trading 212 / InvestEngine APIs	• Investment account linking• Portfolio balance retrieval • Holdings display • Real investment tracking• OAuth authentication	• Link Jordan's Trading 212 account (read-only) • Display actual portfolio vs. scanned projections • Show "You've invested £500 / Scanned £1,247" • Close awareness-to-action loop	Unknown (partnerships required)
ExchangeRate-API	• Currency conversion • Exchange rate retrieval• Multi-currency support • Historical rates • International expansion	• Convert GBP receipts to EUR for European users • Support international stock comparisons • Handle foreign currency purchases • Future: multi-region expansion	Free tier available

Hardware and Sensors

Camera Sensor:

Enables receipt scanning using the device's rear camera. The camera captures clear images of receipts for OCR processing, allowing instant extraction of text, amounts, and merchant details. The guide overlay ensures accurate framing and reduces scan errors.

Face ID / Touch ID (Biometric Sensor):

Used for secure and fast authentication when opening the app. Utilises iOS's Local Authentication framework and Android's BiometricPrompt API. Provides frictionless login without storing sensitive credentials locally.

Processor (CPU/GPU):

Handles on-device OCR and investment projection calculations quickly. Ensures smooth performance during image processing and visual rendering of results.

Storage (Local Device Memory):

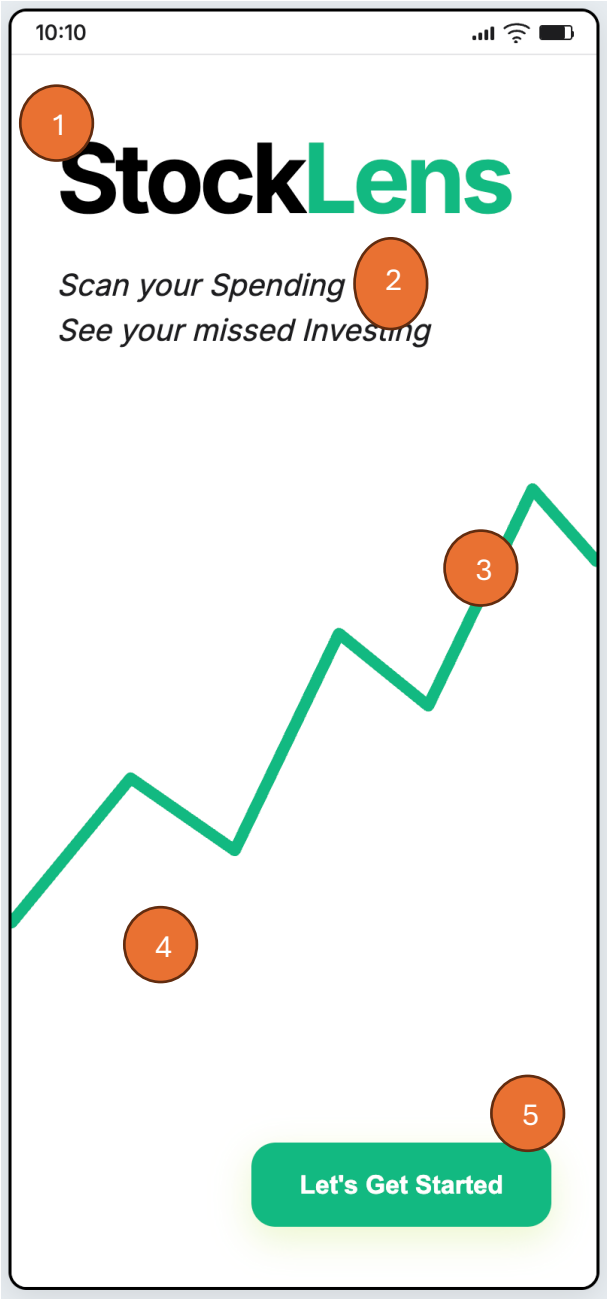
Stores scanned receipt data and projections in an encrypted local SQLite database. No cloud storage is required for MVP, supporting offline use and privacy compliance.

Network Components:

Used only when fetching live or historical stock data via the Alpha Vantage API. Operates securely through HTTPS connections.

Design Mock Ups

1. **Onboarding Screen**



Annotation 1: Two-Tone Logo
Description: Black "Stock" + Green "Lens" (#10B981) creates visual brand identity. Large 32px sizing ensures legibility on small mobile screens during brief splash display.

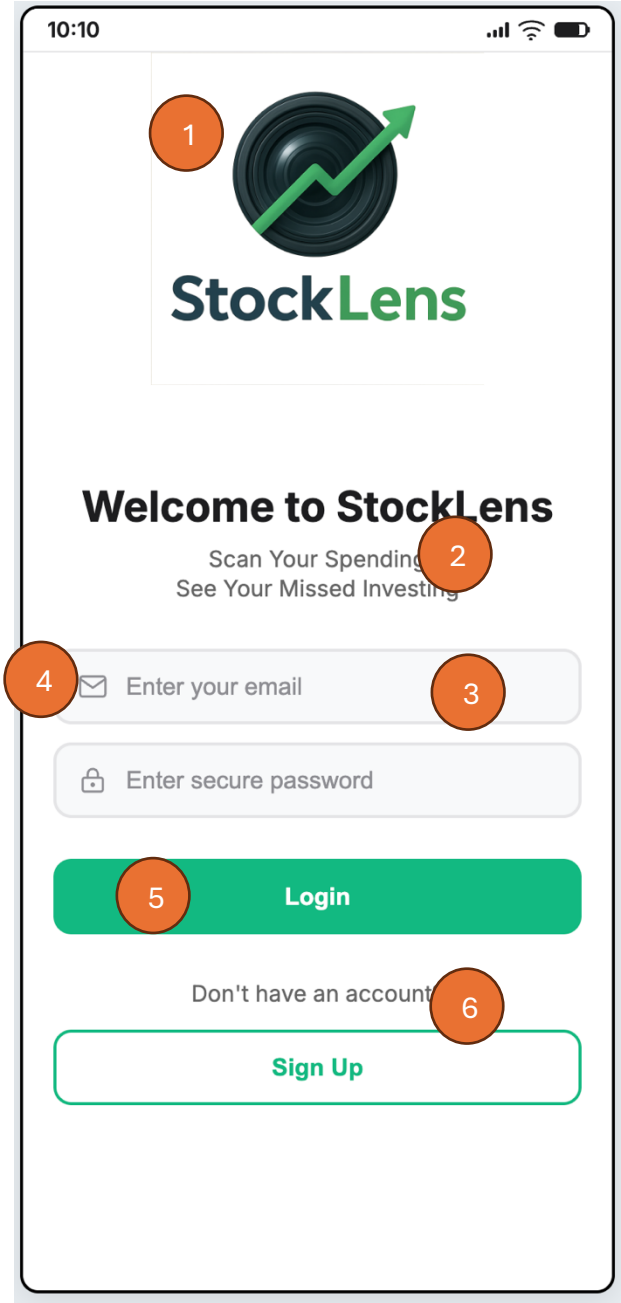
Annotation 2: Dual-Line Value Proposition
Position: Tagline text **Description:** "Scan your Spending / See your missed Investing" uses action-outcome structure. Italic styling adds emphasis. 20px sizing balances readability with hierarchy below logo. Ensures both lines fully fit within 2 lines.

Annotation 3: Growth Chart Visual
Description: Brand green (#10B981) chart communicates stock growth concept without text. Peaks/valleys show realistic market volatility. Positioning keeps it out the way of the logo and the button and nicely isolates the image in the centre for great emphasis.

Annotation 4: White Space Strategy
Description: 60% white space prevents cognitive overload during first impression. Creates professional, uncluttered aesthetic crucial for financial app trust. Improves readability for users with visual processing difficulties.

Annotation 5: Primary CTA Button Design
Description: 350×60px size exceeds 44px minimum accessibility standard for thumb-friendly mobile tapping. Green (#10B981) maintains brand consistency. Pill-shaped (large border radius) feels approachable vs. corporate rectangles. Bottom positioning allows value absorption before commitment. respond 2x stronger to losses than gains). Maintains educational tone without guilt.

2. Login Screen



Annotation 1: Camera Lens Logo with Growth Arrow
Description: Camera lens metaphor with green upward arrow communicates dual concept: scanning (camera) + investment growth (arrow). 120×120px sizing ensures visibility on mobile while maintaining visual balance. Dark lens with brand green arrow creates professional fintech aesthetic.

Annotation 2: Value Proposition Reminder
Description: "Scan Your Spending / See Your Missed Investing" reinforces app purpose before login commitment. 16px grey text (#666666) creates hierarchy below heading. Reminds returning users of value while educating new users who skipped onboarding.

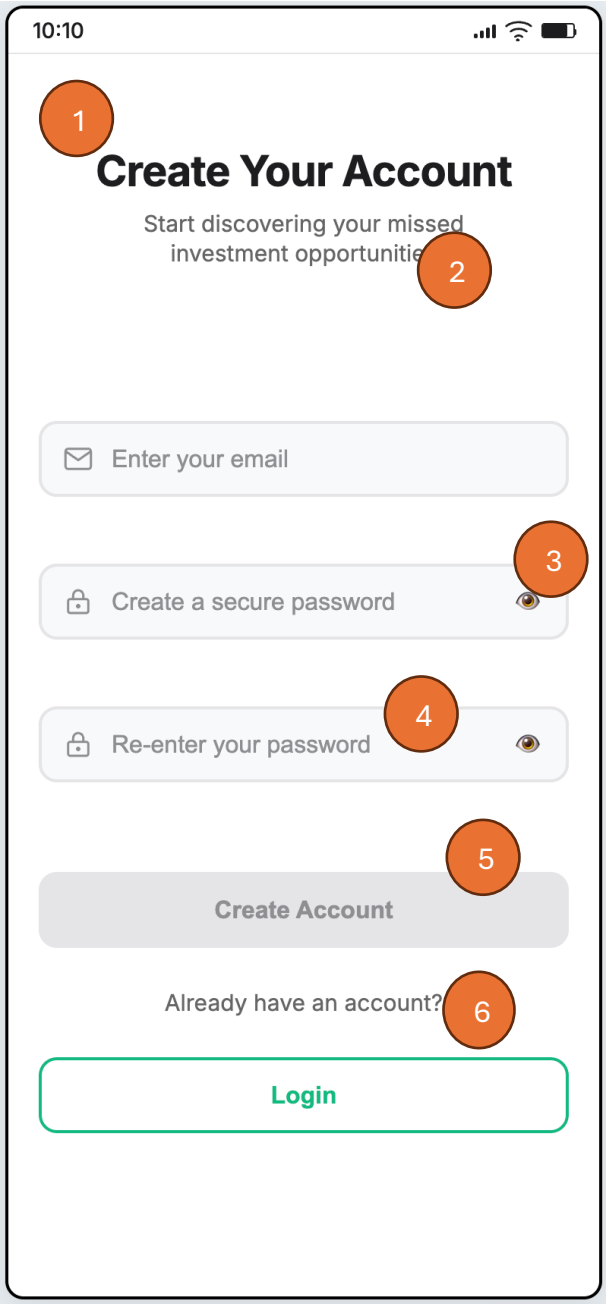
Annotation 3: Input Field Sizing
Position: Email and password fields **Description**: Full-width minus 40px margins maximises tap target on mobile. 56px height exceeds 44px accessibility minimum for comfortable thumb tapping. Light grey background (#F5F5F5) differentiates from white screen while maintaining readability. "Enter your email" and "Enter secure password" placeholders guide user input. "Secure password" reinforces security without enforcing specific requirements that deter sign-ups. 16px sizing matches input text for consistent reading experience.

Annotation 4: Icon Visual Cues
Position: Envelope and lock icons in fields **Description**: Left-aligned icons (24×24px) provide visual field identification before reading labels. Email envelope and password lock are universal symbols reducing cognitive load. Gray colour (#8E8E93) prevents distraction from input text.

Annotation 5: Primary Action Emphasis
Description: Full-width button (56px height) with brand green (#10B981) creates unmissable primary action. White text provides strong contrast. 12px border radius maintains modern iOS design language. Positioned immediately after inputs for logical flow.

Annotation 6: Navigation Question Pattern
Position: "Don't have an account?" text **Description**: Gray text (#666666) at 16px provides context for Sign Up action. Positioned between primary and secondary buttons prevents confusion about which action new users should take. Standard authentication pattern users recognize.

3. Signup Screen



Annotation 1: Action-Oriented Header
Position: "Create Your Account" heading **Description**: 32px bold text uses active verb "Create" establishing user agency. Centred alignment draws focus. Positioned 80px from top provides comfortable reading distance on mobile screens. Black text ensures maximum contrast for readability.

Annotation 2: Motivational Subtitle
Position: Value proposition text below header **Description**: "Start discovering your missed investment opportunities" reinforces benefit before form. 16px grey text (#666666) creates hierarchy below heading. Line-height 1.4 prevents cramping. Motivates completion by reminding users why they're signing up.

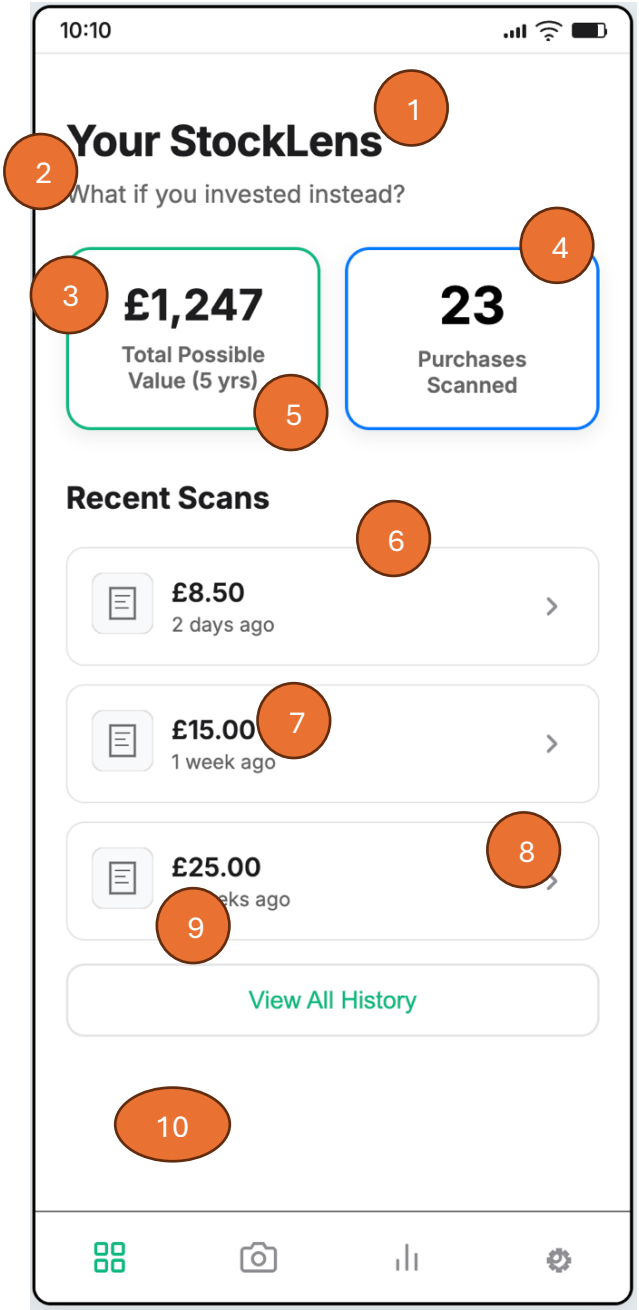
Annotation 3: Password Visibility Toggle
Position: Eye icons on password fields **Description**: Right-aligned eye icons (24×24px) enable show/hide password functionality. Addresses accessibility for users struggling with masked input. Positioned 16px from right edge within comfortable thumb reach. Standard pattern users recognize from other apps.

Annotation 4: Password Confirmation Field
Position: "Re-enter your password" field **Description**: Duplicate password entry prevents typo-related lockouts. Critical for financial app where account recovery is sensitive. 90px spacing from create field provides clear separation. Same styling maintains form consistency.

Annotation 5: Disabled Button State
Position: Greyed out "Create Account" button **Description**: Light grey (#666666) disabled state signals incomplete form. 56px height maintains consistency with other buttons. Full-width design ensures visibility. Prevents premature submission attempts, providing visual feedback about form status. Button only activates when all fields complete, teaching users form requirements through visual feedback rather than error messages.

Annotation 6: Alternative Path Navigation
Position: "Already have an account?" with Login button **Description**: Gray question text (16px) with green outline Login button creates clear secondary path for returning users. 30px spacing separates from primary action. Outline styling differentiates from solid Create Account button when enabled.

4. Dashboard Screen



Annotation 1: Personalized Header
Position: "Your StockLens" heading **Description**: 32px bold text with possessive "“Your" creates personal ownership of financial data. Left-aligned for mobile readability. 40px top padding provides comfortable viewing distance from status bar.

Annotation 2: Thought-Provoking Tagline
Position: "What if you invested instead?" subtitle **Description**: 16px grey text (#666666) poses reflective question reinforcing app's core concept. Positioned directly below header with 8px spacing. Frames dashboard metrics as opportunity cost rather than judgment.

Annotation 3: Primary Metric Emphasis
Position: Left green card with £1,247 **Description**: Green border (#10B981) frames positive potential value. 48px bold number creates immediate visual impact. "Total Possible Value (5 yrs)" uses specific timeframe for credibility. Card positioning left-first follows reading flow.

Annotation 4: Engagement Tracking Card
Position: Right blue card with 23 **Description**: Blue border differentiates from monetary metric. 48px bold "23" celebrates user activity. "Purchases Scanned" reinforces behaviour rather than loss. Provides positive reinforcement balancing opportunity cost messaging.

Annotation 5: Recent Scans Section Header
Position: "Recent Scans" text **Description**: 24px bold black text creates clear section break. Left-aligned with 30px top margin from summary cards. Establishes content hierarchy for scrollable list below.

Annotation 6: Receipt List Item Design
Position: Individual scan cards **Description**: Each card 80px height provides comfortable tap target for viewing details. Receipt icon (40×40px) in grey square creates consistent visual anchor. Light grey background (#F8F9FA) with subtle border separates items. 12px vertical spacing prevents accidental taps.

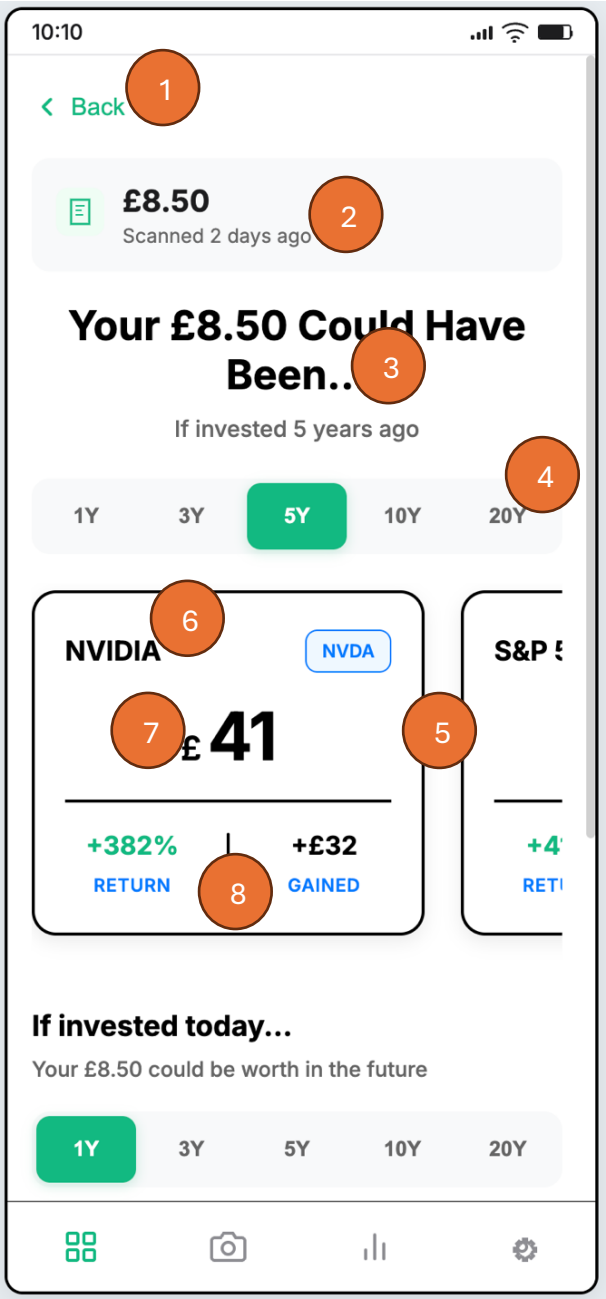
Annotation 7: Temporal Context Display
Position: Timestamps under amounts **Description**: "2 days ago" relative time format reduces cognitive load vs absolute dates. 14px grey text creates hierarchy below 20px black amount. Helps users identify recent vs. older spending patterns.

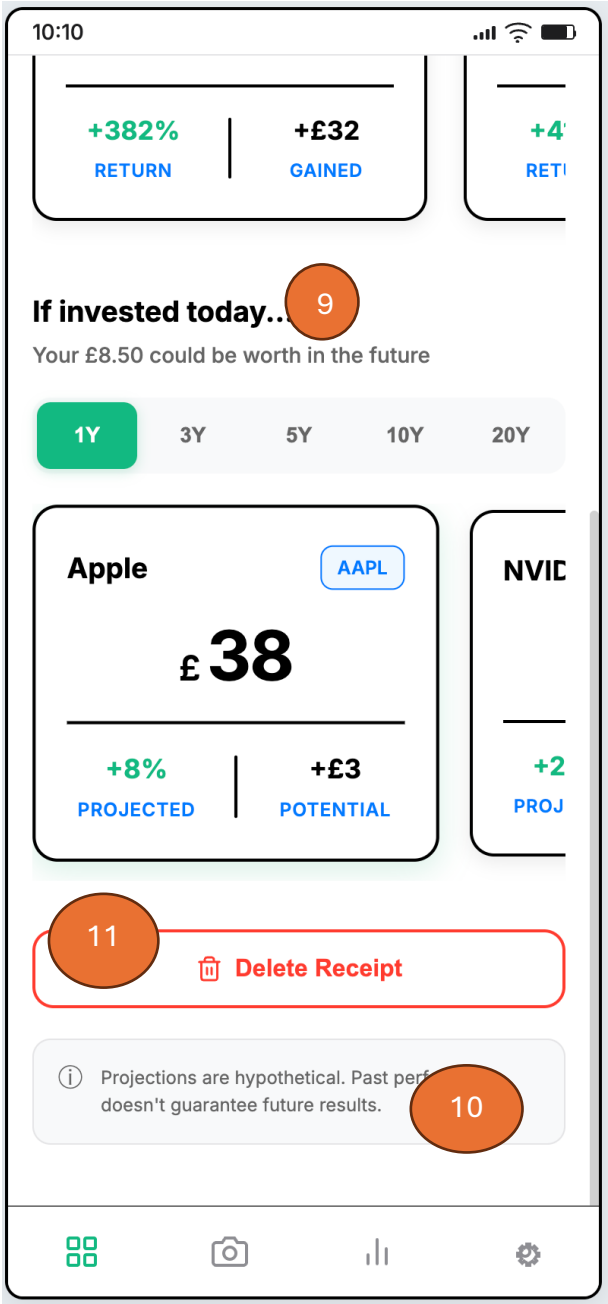
Annotation 8: Chevron Disclosure Indicator
Position: Right-aligned arrows on list items **Description**: Right arrows (20×20px, #C7C7CC) signal tappable items leading to detail views. Standard iOS pattern users recognize. Positioned 16px from right edge within thumb reach. Indicates deeper content without cluttering interface.

Annotation 9: Progressive Disclosure Button
Position: "View All History" button **Description**: Full-width button (48px height) with green text provides clear path to complete data. Border-only styling signals secondary action. 20px top margin separates from last list item. Accommodates users wanting comprehensive view.

Annotation 10: Tab Bar Navigation
Position: Bottom navigation bar **Description**: 80px height accommodates safe area on notched phones. Four evenly spaced tabs support primary app functions. Dashboard icon (green, active state) provides spatial orientation. Icon sizing 28×28px ensures visibility and tap-ability.

5. Receipt Detail Screen





Annotation 1: Hierarchical Back Navigation
Position: Green "Back" text with chevron **Description**: 44×44px tap target meets accessibility minimum. Green colour (#10B981) maintains brand consistency. Left-aligned follows iOS convention for navigation. Positioned 20px from top for comfortable thumb reach.

Annotation 2: Scan Context Card
Position: Receipt icon with £8.50 and timestamp **Description**: Light grey card (40px height) provides context for projections below. Receipt icon (32×32px) with green accent reinforces scanning action. "Scanned 2 days ago" temporal context helps users identify transaction. 16px padding creates breathing room.

Annotation 3: Emotional Impact Headline
Position: "Your £8.50 Could Have Been..." text **Description**: 28px bold text creates emotional hook using original spend amount. "Could Have Been" language leverages loss aversion psychology. Ellipsis creates curiosity gap encouraging scroll engagement. Centred for emphasis.

Annotation 4: Time Period Segmented Control
Position: 1Y/3Y/5Y/10Y/20Y selector **Description**: Five equal segments (each ~70px wide) showing multiple timeframes simultaneously. 5Y active state (green fill) indicates current view. 36px height provides comfortable mobile tapping. Enables users to compare different investment horizons addressing core app functionality.

Annotation 5: Horizontal Stock Card Scroll
Position: NVIDIA and S&P 500 cards side-by-side **Description**: Cards 280px wide enabling side-by-side comparison with horizontal scroll. Rounded corners (16px radius) and borders create distinct containers. Allows multiple stock benchmark comparisons without vertical clutter. Swipe able pattern users recognize from app stores.

Annotation 6: Stock Identification Design
Position: Company name with ticker badge **Description**: Bold company name (20px) with blue outlined ticker badge (NVDA/AAPL) provides dual identification. Ticker positioned top-right where experienced investors expect it. Badge styling differentiates from primary content while maintaining visibility.

Annotation 7: Projection Value Hierarchy
Position: Large £41/£38 amounts **Description**: 72px bold numbers create immediate visual impact showing potential value. Black text ensures maximum contrast. Size hierarchy communicates "this is what matters most" - the missed opportunity amount. Centred in card for emphasis.

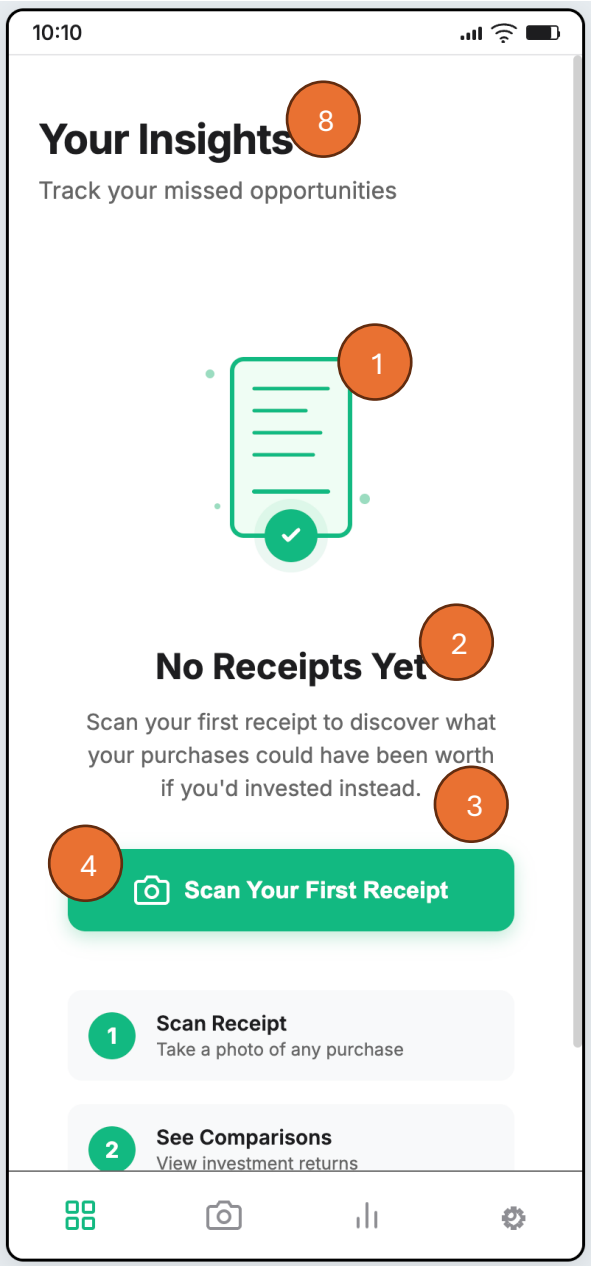
Annotation 8: Dual Metric Display
Position: Return % and Gained £ split **Description**: Green +382% and +£32 displayed side-by-side with vertical divider. Accommodates different user preferences - some think in percentages, others in currency. "RETURN" vs "GAINED" labels (12px blue caps) clarify metric meaning. Equal visual weight signals both metrics matter.

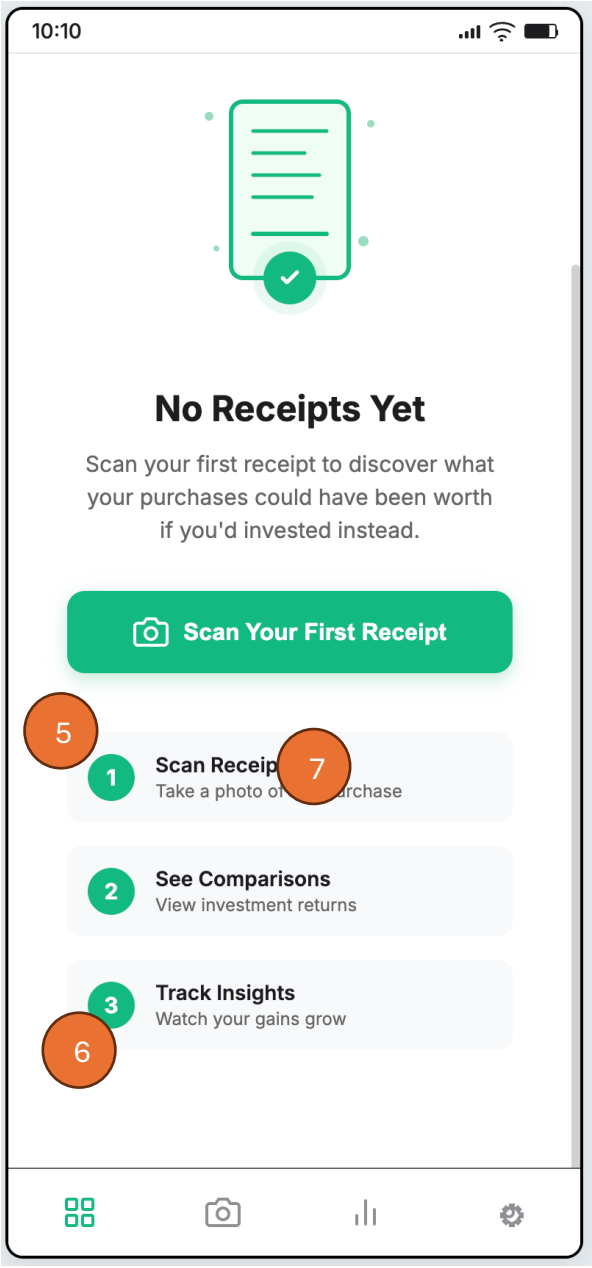
Annotation 9: Future Projection Section
Position: "If invested today..." heading **Description**: 24px bold text creates clear section break. Gray subtitle explains forward-looking nature. 1Y green active state shows shortest projection by default. Different styling from historical section above separates past performance from future projections.

Annotation 10: Legal Disclaimer
Position: Bottom info box with warning text **Description**: Light grey background (#F8F9FA) with info icon distinguishes from actionable content. "Projections are hypothetical. Past performance doesn't guarantee future results." meets regulatory requirements for financial apps. 14px text ensures readability while not dominating interface. Addresses ethical considerations from security section.

Annotation 11: Delete Action Design
Position: Red "Delete Receipt" button **Description**: Full-width button (52px height) with red colour (#FF3B30) and trash icon signals destructive action. Outline styling vs. solid fill provides visual warning before irreversible deletion. Positioned after all content ensures deliberate action rather than accidental tap.

6. Dashboard Empty Screen





Annotation 1: Empty State Illustration Position: Receipt icon with checkmark **Description**: 150×180px illustration uses brand green (#10B981) maintaining visual consistency. Checkmark badge (60×60px) suggests completion/readiness state. Decorative dots create friendly, approachable feel. Positioned centre-screen at eye level for immediate recognition of empty state.

Annotation 2: Clear Empty State Messaging
Position: "No Receipts Yet" heading **Description**: 32px bold text directly communicates current state without technical jargon. Positioned below illustration with 40px spacing. Black text ensures readability. Short, declarative phrase prevents user confusion about why screen appears empty.

Annotation 3: Educational Value Proposition
Position: Three-line explanation text **Description**: 16px grey text (#666666) explains what happens after first scan. "Discover what your purchases could have been worth" reinforces core value before

user commits to action. 280px max width improves readability. Line-height 1.5 prevents cramping on small screens.

Annotation 4: Primary Call-to-Action
Position: Green "Scan Your First Receipt" button **Description**: Full-width minus 40px margins, 56px height creates prominent action target. Camera icon (24×24px) paired with text clarifies button function. Green fill (#10B981) signals primary action. 50px spacing from explanation text prevents accidental taps while scrolling.

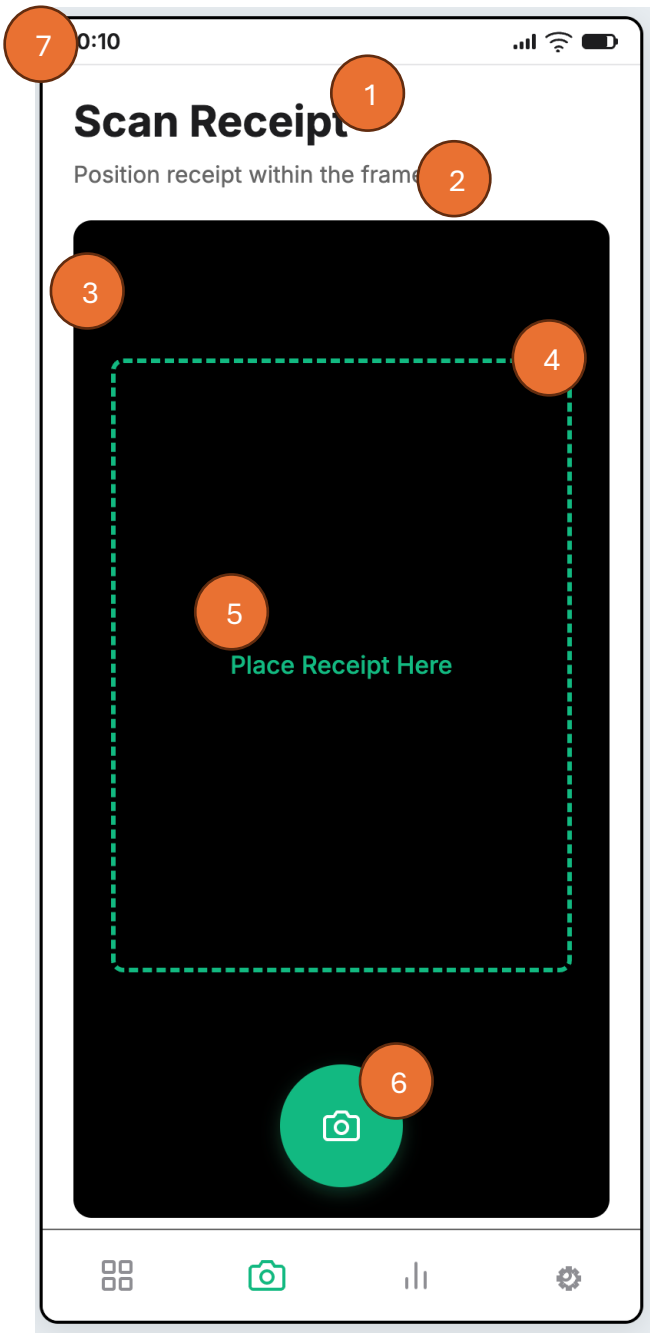
Annotation 5: Step-by-Step Guidance
Position: Numbered instruction list (1-2-3) **Description**: Three steps with 44×44px numbered circles provide clear process overview. Green circles maintain brand consistency. Reduces uncertainty about app functionality. 40px vertical spacing between steps ensures comfortable readability and prevents visual crowding on mobile. Instructions positioned after CTA button allows motivated users to act immediately while providing guidance for uncertain users. Follows natural scroll behaviour. Prevents overwhelming first-time users with information before they're ready to engage.

Annotation 6: Numbered Circle Design
Position: Green circles with white numbers **Description**: 44×44px sizing ensures visibility while maintaining proportion. White numbers on green background provide strong contrast (7:1 ratio). Circle shape universally represents steps/progress. Left-aligned creates scannable vertical rhythm.

Annotation 7: Instruction Text Hierarchy
Position: Bold titles with grey subtitles **Description**: 18px bold titles ("Scan Receipt", "See Comparisons") provide quick scanning. 14px grey subtitles offer additional context without cluttering. Two-tier approach accommodates different reading depths - some users read everything; others scan bold text only.

Annotation 8: Screen Header Consistency
Position: "Your Insights" title (second image) **Description**: 32px bold heading with 16px grey subtitle maintains template consistency across dashboard variations. "Your" creates personal ownership. "Track your missed opportunities" frames purpose positively. Left-aligned for mobile reading patterns. Same positioning as dashboard populated state creates predictable navigation experience.

7. Receipt Scan Screen



Annotation 1: Task-Focused Header
Position: "Scan Receipt" title Description: 36px bold text uses imperative verb establishing clear action. Left-aligned for mobile reading patterns. 40px top padding provides comfortable distance from status bar. Direct language reduces confusion about screen purpose.

Annotation 2: Instructional Guidance
Position: "Position receipt within the frame" subtitle Description: 16px grey text (#666666) provides specific positioning instructions. Reduces user anxiety about camera usage. Positioned 12px below header creates clear hierarchy. Plain language accessible to all users regardless of technical literacy.

Annotation 3: Full-Screen Camera Viewport
Position: Black camera preview area Description: Occupies maximum vertical space (approximately 85% of screen height) for optimal receipt framing. Black background simulates real camera viewfinder experience users recognize from native camera apps. 12px border radius maintains design system consistency.

Annotation 4: Receipt Guide Overlay
Position: Dashed green rectangle Description: 540×720px (portrait receipt ratio 3:4) accommodates standard receipt dimensions. 4px dashed border (#10B981) provides clear targeting without obstructing camera view. Dashed pattern suggests temporary guide that won't appear in final capture. Centred positioning supports natural camera holding angle.

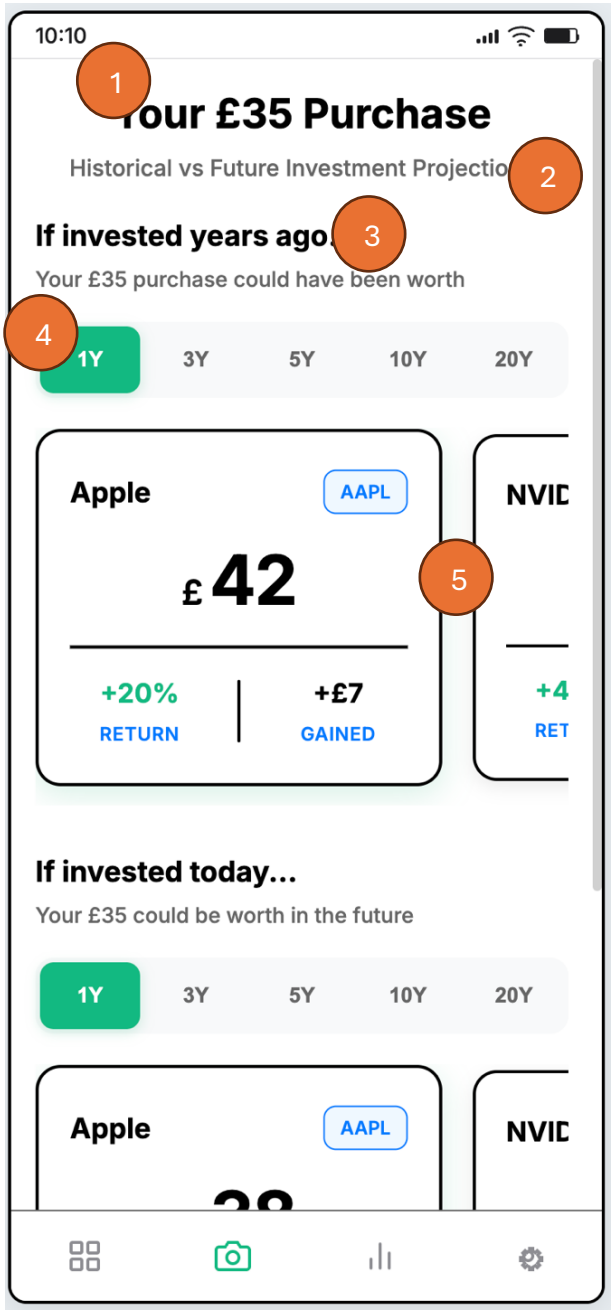
Annotation 5: Centred Helper Text
Position: "Place Receipt Here" text in frame Description: 20px medium green text positioned centre of guide frame. Reinforces positioning instruction visually within target area. Green colour maintains brand consistency while standing out against black camera background. Disappears when receipt detected (in final implementation).

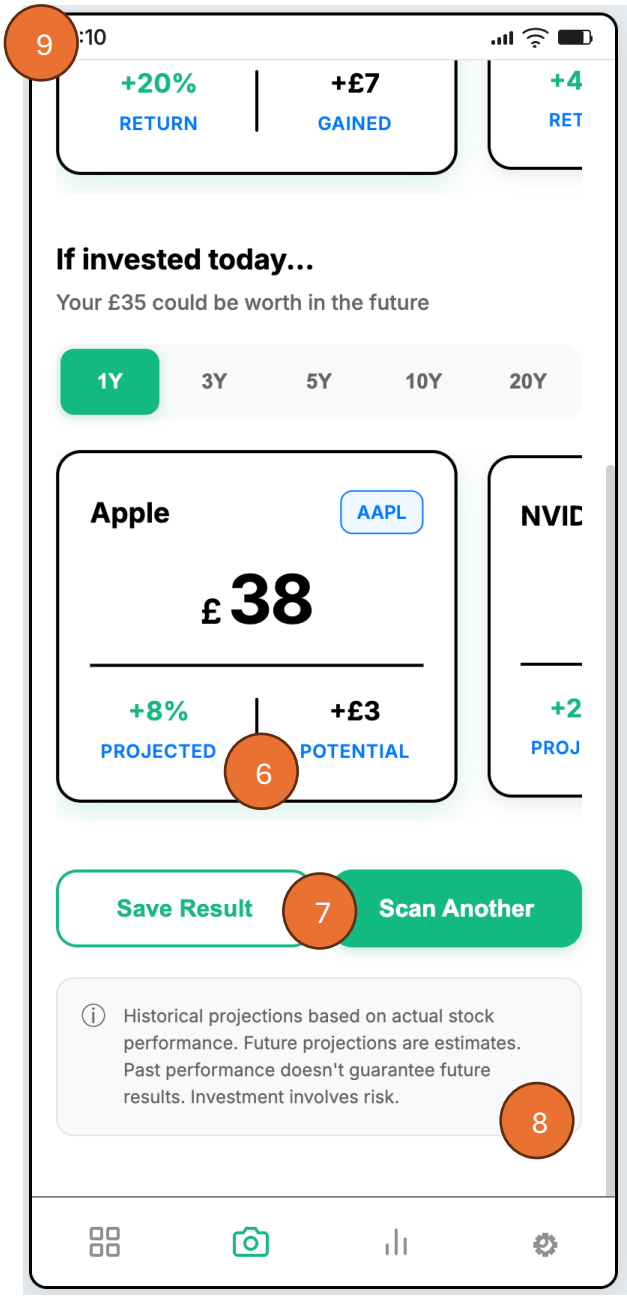
Annotation 6: Floating Capture Button
Position: Circular green button at bottom Description: 80×80px circular button mirrors native iOS camera shutter design leveraging learned behaviour. Green background (#10B981) with white camera icon (32×32px) creates strong contrast. Positioned 60px from bottom accommodates thumb reach on large phones. Drop shadow creates floating effect suggesting primary action. White camera

symbol in button, 32×32px icon size ensures visibility within 80px button. White colour provides 7:1 contrast ratio against green background. Simple camera glyph universally recognized across cultures and age groups. Icon-only approach keeps interface clean without competing text labels.

Annotation 8: Minimalist Camera Interface
Position: Overall screen design **Description**: No flash toggle, zoom controls, or settings visible keeps interface focused on single task: capturing receipt. Reduces decision paralysis and accidental setting changes. Addresses Jordan's short attention span and Aisha's preference for simple interfaces. Advanced controls accessible via settings if needed.

8. Receipt Scan Results Screen





Annotation 1: Contextualized Header
Position: "Your £35 Purchase" title Description: 32px bold text personalizes results by referencing exact scanned amount. "Your" creates ownership. Positioned 40px from top for comfortable reading. Black text ensures maximum contrast for quick comprehension.

Annotation 2: Dual Comparison Framework
Position: "Historical vs Future Investment Projections" subtitle Description: 16px grey text (#666666) establishes two-part structure users should expect below. Prepares users mentally for scrollable content comparing past performance vs. future estimates. Transparent framing builds educational credibility.

Annotation 3: Section Divider Headers
Position: "If invested years ago..." and "If invested today..." text Description: 24px bold section headers create clear content separation. "Years ago" establishes historical lookback, "today" frames forward

projection. Positioned with 30px top padding provides visual breathing room between scrollable card sections.

Annotation 4: Time Period Pill Selection
Position: 1Y/3Y/5Y/10Y/20Y segmented controls Description: Five equal-width segments (each ~70px) enable comparative analysis across timeframes. 1Y active state (green fill) shows shortest historical period by default. 40px height provides comfortable mobile tapping. Users can quickly switch perspectives without leaving screen.

Annotation 5: Horizontal Stock Card Scroll
Position: Side-by-side Apple and NVIDIA cards Description: 280px card width with 15px gap enables side-by-side comparison while suggesting more cards exist off-screen. Swipe able horizontal scroll pattern familiar from App Store and social media. Maintains vertical screen space for dual historical/future sections.

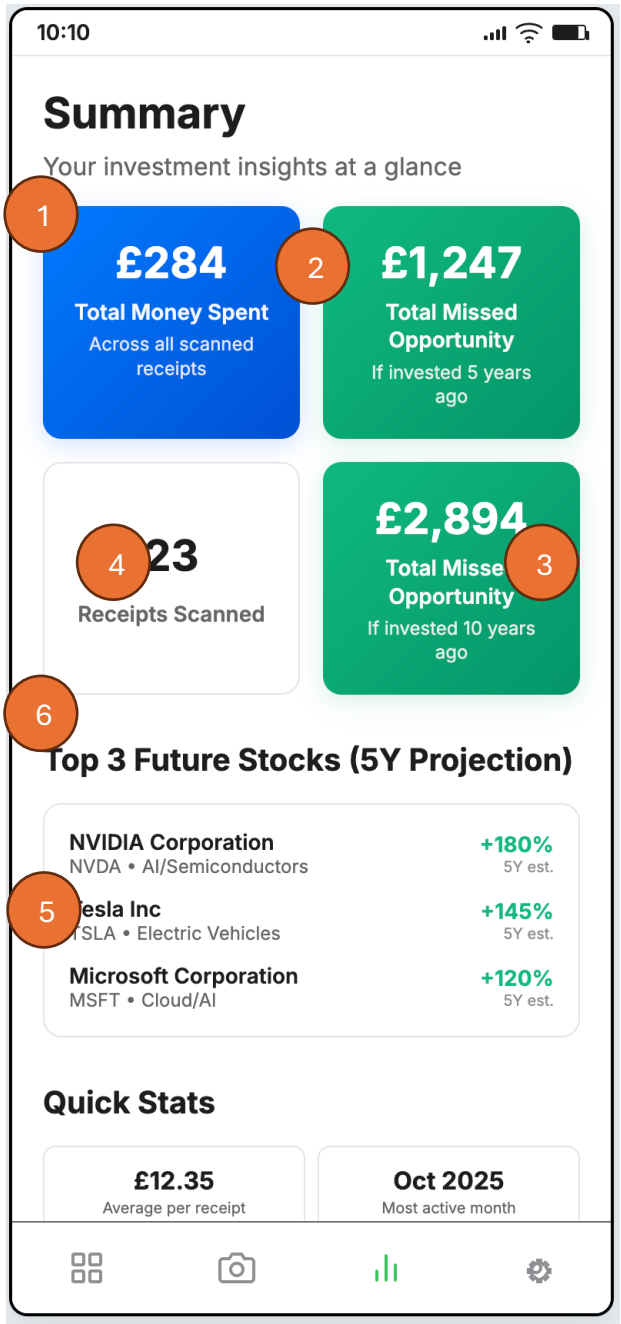
Annotation 6: Metric Label Differentiation
Position: "RETURN/GAINED" vs "PROJECTED/POTENTIAL" labels Description: 12px blue uppercase labels distinguish historical facts (RETURN/GAINED) from future estimates (PROJECTED/POTENTIAL). Word choice signals certainty level - past is concrete, future is hypothetical. Addresses regulatory requirements for financial projections.

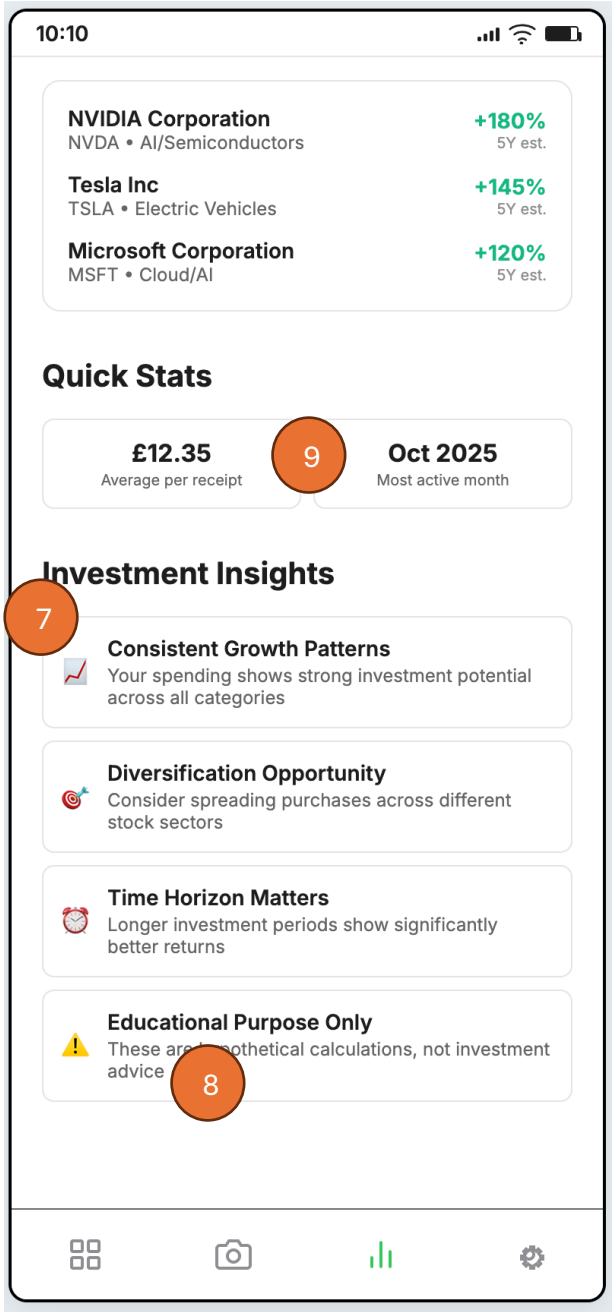
Annotation 7: Action Button Pair
Position: "Save Result" and "Scan Another" buttons Description: Equal-width buttons (48% each with 4% gap) provide balanced action choices. "Save Result" outline styling signals secondary action, "Scan Another" solid green emphasizes continued engagement. Side-by-side layout accommodates different user intentions without hierarchy.

Annotation 8: Comprehensive Disclaimer
Position: Light grey info box at bottom Description: Light background (#F8F9FA) with info icon differentiates legal text from actionable content. Four-sentence disclosure covers historical accuracy, future uncertainty, past performance limitations, and risk acknowledgment. 14px text size balances legal necessity with readability. Meets financial app regulatory requirements while maintaining user-friendly presentation.

Annotation 9: Scrollable Dual-Section Layout
Position: Overall screen organization Description: Vertical scroll accommodates two complete projection sections (historical + future) without cramping. Each section gets equivalent screen real estate signalling equal importance. Addresses core app value proposition of showing both retrospective and prospective investment scenarios.

9. Summary/Info Screen





Annotation 1: Summary Card Grid Layout
Position: Four-card grid at top **Description**: 2×2 grid layout (two rows, two columns) maximises mobile screen efficiency. Blue card (£284 spent) uses contrasting colour differentiating from green opportunity cards. Equal 160×180px card sizing creates visual balance. 15px gaps prevent cramping while maintaining scannable density.

Annotation 2: Color-Coded Metric Cards
Position: Blue vs green cards **Description**: Blue (#007AFF) for factual spending, green (#10B981) for opportunity cost creates immediate visual categorisation. Colour psychology: blue = information/trust, green = growth/money. Helps users distinguish between "what happened" (blue) vs "what could have been" (green).

Annotation 3: Hierarchical Typography in Cards
Position: Large amounts with smaller context text **Description**: 48px bold white numbers create immediate impact. 18px subtitle text provides context without competing. "Across all scanned receipts" and "If invested X years ago" clarifies metric meaning. Three-tier hierarchy (amount → label → context) accommodates quick scanning and detailed reading.

Annotation 4: Neutral Engagement Metric
Position: White card with "23 Receipts Scanned" **Description**: White background with black text provides neutral presentation for activity metric. Positioned bottom-left balances blue spending card above. Celebrates engagement without framing as loss. 72px number sizing maintains consistency with other cards.

Annotation 5: Top Performers List
Position: Stock ranking with performance percentages **Description**: Three-item list shows best-performing stocks without overwhelming with choices. Company name (18px bold) + ticker/sector (14px grey) provides dual identification. Right-aligned green percentages create scannable column. Light grey card background separates from white screen.

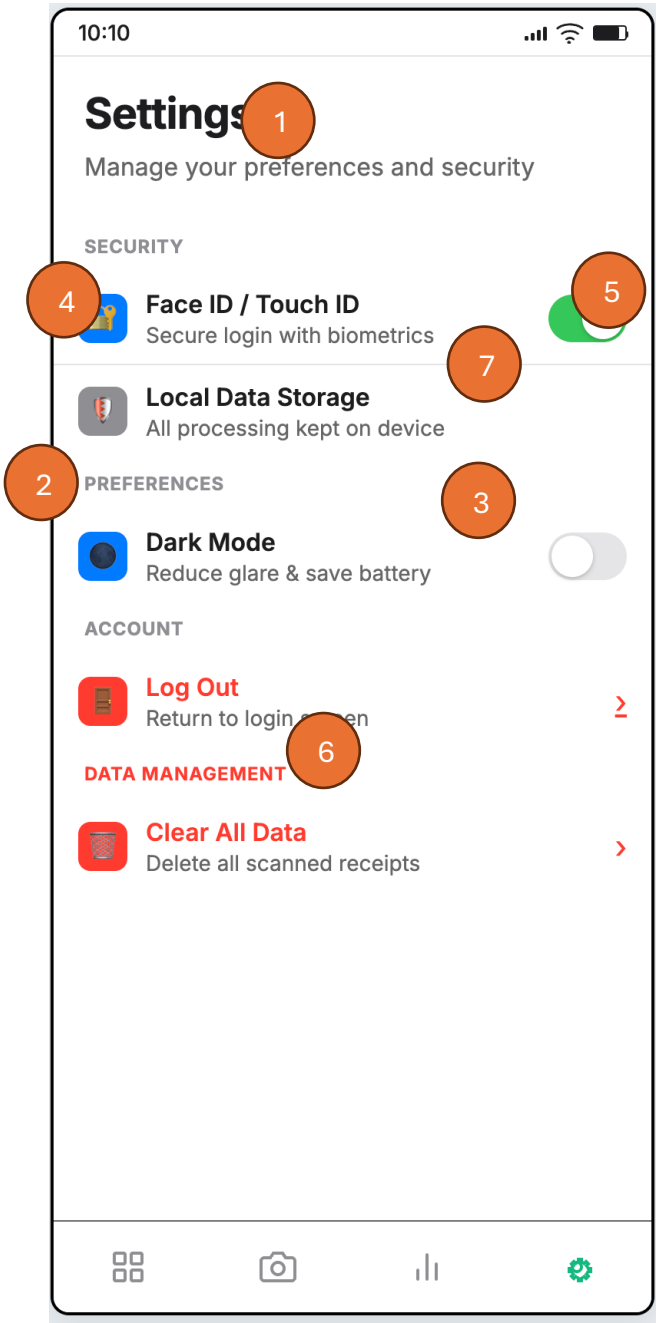
Annotation 6: Section Header Clarity
Position: "Top 3 Future Stocks (5Y Projection)" heading **Description**: 24px bold heading with specific timeframe "(5Y Projection)" manages expectations about data shown. Clarifies forward-looking nature vs historical performance. Positioned with 30px top spacing provides visual section break in scrollable content.

Annotation 7: Insight Card Design
Position: Educational cards with icons and text **Description**: Each card 80px height with 16px padding provides comfortable reading space. Left-aligned emoji icons (32×32px) create visual anchors and emotional associations. Bold 18px titles with 14px grey body text creates hierarchy. Cards use light backgrounds differentiating insight types.

Annotation 8: Warning Disclaimer Styling
Position: Yellow warning icon with "Educational Purpose Only" **Description**: Yellow warning triangle (#FFD60A) leverages universal caution symbolism. "Educational Purpose Only" bold title with explanatory text addresses regulatory requirements. Positioned last ensures users see insights before limitations. Light yellow background (#FFF9E6) draws attention without alarming.

Annotation 9: Quick Stats Metrics
Position: Two-column stat display **Description**: Side-by-side layout shows complementary metrics: average (£12.35) and temporal pattern (Oct 2025). 28px bold numbers with 14px grey labels. Equal card widths signal balanced importance. Provides quick context without requiring detailed analysis.

10. Settings Screen



Annotation 1: Clear Settings Header
Position: "Settings" title with subtitle **Description**: 36px bold title establishes screen purpose. 16px grey subtitle "Manage your preferences and security" sets expectations for content below. Left-aligned for mobile readability. 40px top padding provides comfortable viewing distance from status bar.

Annotation 2: Section Header Styling
Position: "SECURITY", "PREFERENCES", "ACCOUNT" labels **Description**: 12px uppercase grey text (#8E8E93) creates visual separation between setting groups. All-caps with letter-spacing follows iOS convention for section headers. 30px top margin from previous section provides clear content breaks in scrollable list. Four distinct sections, Security first reflects priority. Preferences for customization. Account for authentication. Data Management separated emphasizes destructive nature. Logical flow from protection → personalization → access → deletion follows mental model of increasing consequence.

Annotation 3: Setting Item Height
Position: All individual setting rows **Description**: 60px minimum height ensures comfortable tap targets exceeding 44px accessibility standard. Accommodates two-line layout (title + description) without cramping. Horizontal padding 16px provides thumb-friendly margins on both edges.

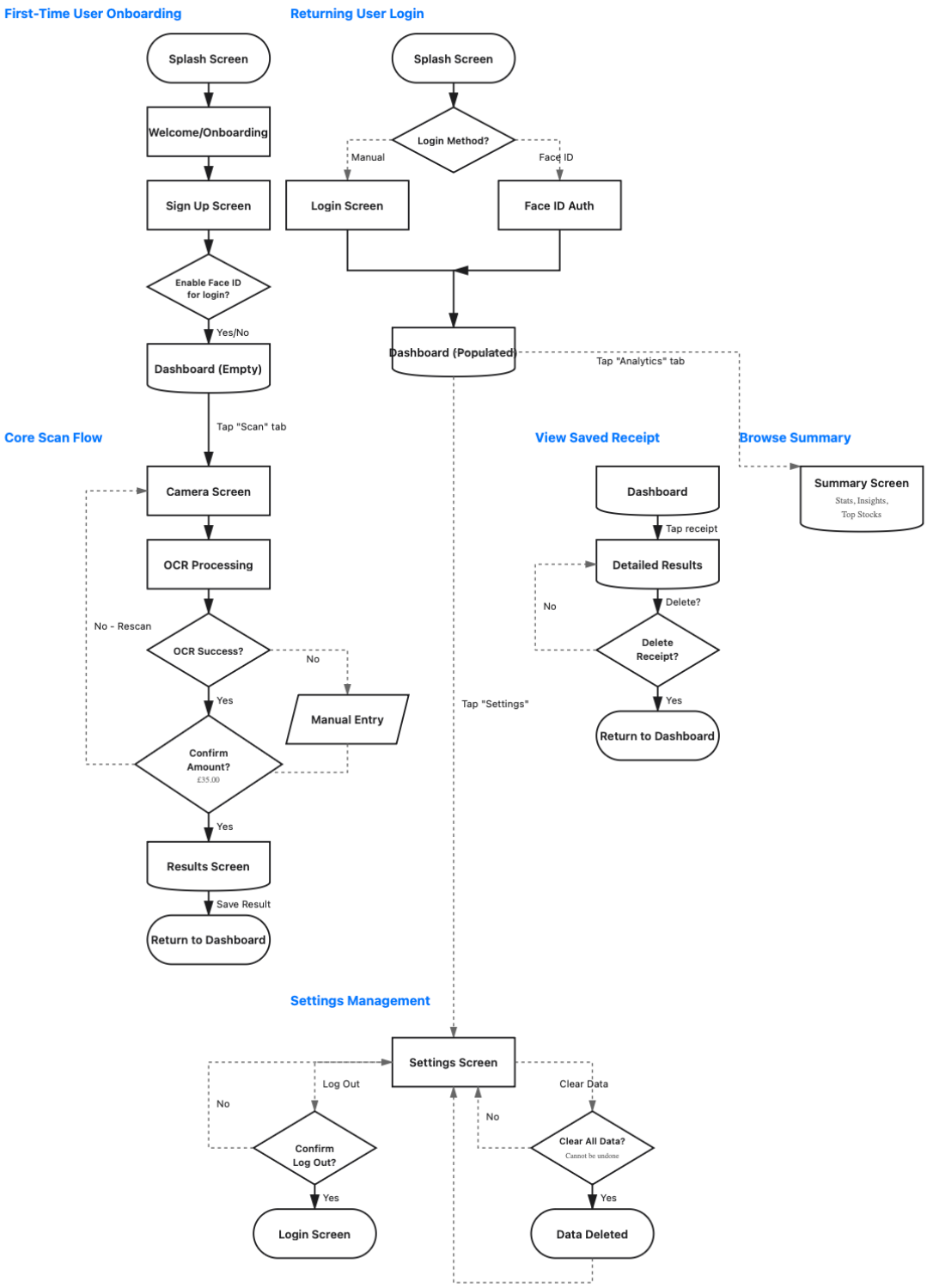
Annotation 4: Icon Visual System
Position: Coloured square icons (40×40px) **Description**: 40×40px icons with 8px border radius create consistent visual anchors. Colour coding: blue (features), grey (info), red (destructive). Lock, shield, moon, and trash symbols universally recognizable. Positioned left with 12px margin from text.

Annotation 5: Toggle Switch Design
Position: Face ID and Dark Mode switches **Description**: 51×31px iOS-standard toggle size ensures familiarity. Green ON state (#10B981) maintains brand consistency. Gray OFF state signals inactive. Right-aligned positioning follows iOS convention. Provides instant feedback without navigation.

Annotation 7: Destructive Action Styling
Position: Red "Log Out" and "Clear All Data" items **Description**: Red text (#FF3B30) and icons signal irreversible actions. Right chevron arrows indicate navigation to confirmation screens preventing accidental deletion. Red section headers emphasize danger zone. Positioned after regular settings as warning pattern.

Annotation 8: Privacy-First Messaging
Position: "Local Data Storage" item **Description**: Shield icon with "All processing kept on device" communicates privacy commitment addressing security section requirements. No toggle suggests always-on privacy rather than optional feature. Gray icon indicates informational rather than actionable item.

User Flow Diagram



Navigation Structure

Bottom Tab Bar (Always Available):
- Dashboard (Home icon)
- Camera (Scan icon)
- Analytics (Chart icon)
- Settings (Gear icon)

Back arrows return to Dashboard
Popups have Yes/No options

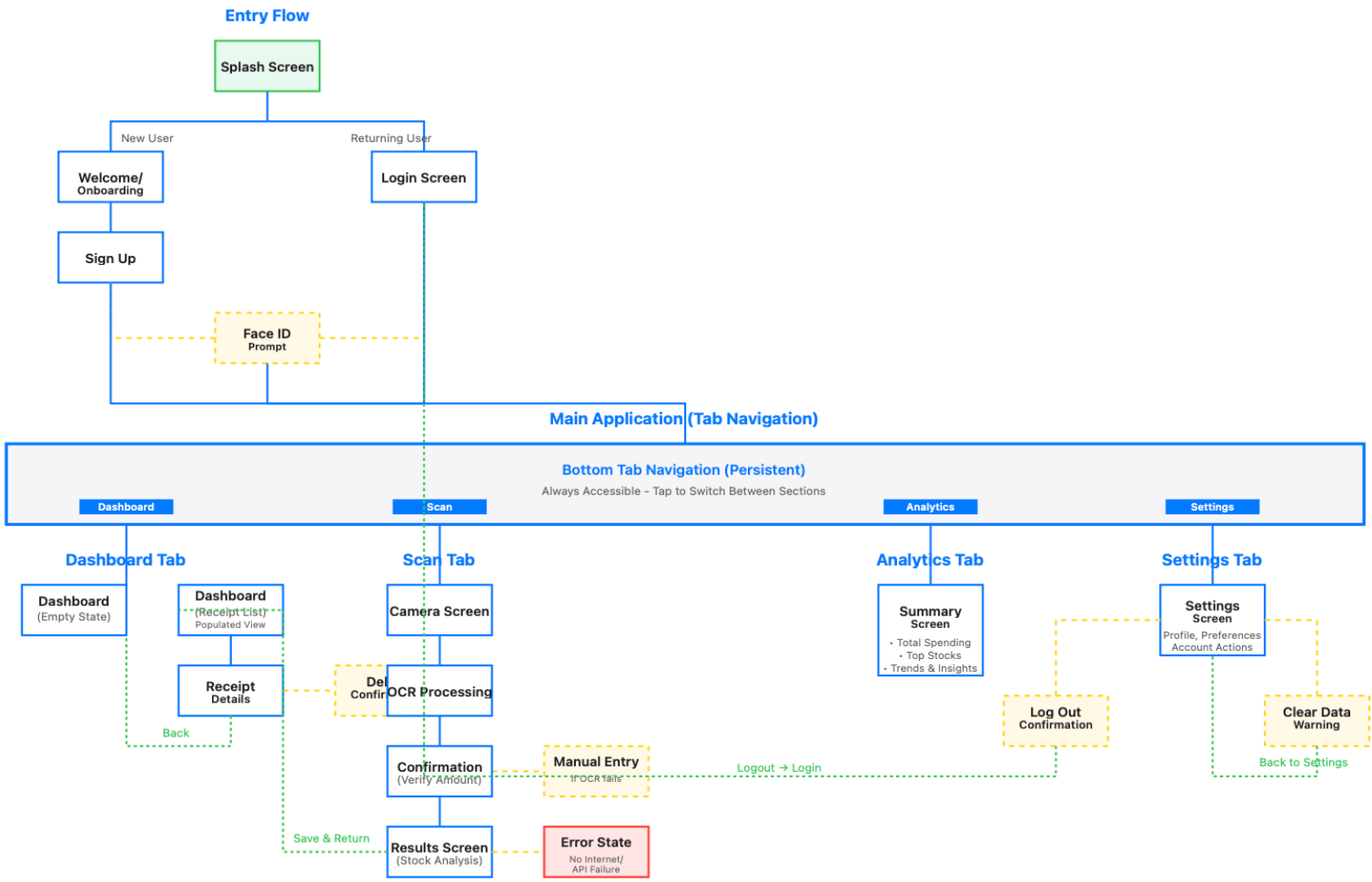
Error Handling

- OCR Failure → Show "Unable to read receipt" → Manual entry option
- API Failure → Show cached data with "Using estimated data" disclaimer
- No Internet → "No connection" message → Retry OR use cached data

Flowchart Symbol Legend

- Terminator (Start/End)
- Process
- Decision (Yes/No)
- Manual Input
- Display/Output
- Primary Flow
- Alternative Path

App Map



Legend

- Entry Point (Splash, Auth)
- Standard Screen
- Modal/Popup/Confirmation
- Error/Warning State
- Main Navigation Container
- Forward Navigation
- Modal Trigger
- Back/Return Path

App Structure Notes

Navigation Model:
- Bottom Tab Bar is persistent across all main sections
- 4 primary tabs: Dashboard, Scan, Analytics, Settings
- Each tab maintains its own navigation stack
- Green dashed arrows show return/back navigation paths

User States & Flows:
- New Users: Splash → Welcome → Sign Up → Face ID → Dashboard (Empty)
- Returning Users: Splash → Login/Face ID → Dashboard (Populated)
- Dashboard Empty State prompts first scan action

Error Handling & Edge Cases:
- OCR Failure → Manual Entry modal appears as fallback
- API/Network Error → Error state shown with cached data option
- All destructive actions (Delete, Clear Data, Logout) require confirmation

Screen Count Summary

- Total Unique Screens: **18**
- Main Content Screens: **12**
- Modal/Popup Screens: **5**
- Error/Warning States: **1**
- Entry/Auth Screens: **4**

Key Design Decisions

- **Flat hierarchy:** Max 3 levels deep from any tab
- **Quick access:** Core scan function always 1 tap away
- **Context preservation:** Tab bar always visible for quick switching between major sections
- **Error resilience:** Fallback options for OCR failure and offline scenarios with cached data
- **State awareness:** Different empty states guide new users vs showing data for returning users

Security and Accessibility Considerations

Security Implementation

StockLens uses a privacy-by-design architecture where receipt images undergo on-device OCR processing and are only saved on device (not the cloud), so no sensitive purchase data transmits externally, ensuring GDPR compliance and eliminating data breach risks.

Firebase Authentication enforces industry-standard password hashing (bcrypt) with biometric authentication (Face ID/Touch ID) via platform-native APIs, providing secure, frictionless login without storing credentials locally. SQLite database employs AES-256 encryption protecting stored receipt history from unauthorized device access. Alpha Vantage API calls utilize HTTPS/TLS 1.3 encryption with environment-secured API keys, never exposed in client code. All financial calculations occur locally using cached historical data, minimizing external dependencies and enhancing offline security.

Accessibility Standards

Full WCAG 2.1 AA compliance ensures inclusive design: minimum 4.5:1 colour contrast ratio accommodate colour vision deficiencies, while 48×48px touch targets exceed accessibility guidelines for motor-impaired users.

Ethical Safeguards

Prominent multi-screen disclaimers clarify projections are hypothetical educational tools, explicitly stating "not FCA-regulated investment advice" to prevent vulnerable users from uninformed financial decisions. Zero dark patterns, no artificial urgency tactics, no sponsored investment platform recommendations, no affiliate links promoting specific brokerages. Age-appropriate content ensures 16-17-year-old secondary users (David's students) receive suitable financial literacy education without inappropriate risk exposure or investment solicitation.